MACKING SCHEME

PHYSICS 1

F 4 EXAM 2024

1. (10marks)

(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)	(x)
C	D	A	B	E	E	E	C	C	B

2. (5marks)

LIST A	(i)	(ii)	(iii)	(iv)	(v)
LIST B	C	A	B	E	D

3. (a) (i) By Letting off some air from the tyres, each tyre flattens there by increasing the surface area in contact hence the pressure exerted by each tyre on the soft sand is reduced. In so doing prevents ~~they~~ tyres from sinking. (4MKs)

(ii) Atmospheric pressure is exerted on both sides of a window so no net force is exerted on the window and hence uniform pressure does not break glass windows. (4marks)

3 (b) Data:

Mass of a trolley = 0·5kg

Speed of a trolley, V = 2m/s

From; The principle of Conservation of energy:

Maximum P.E = Maximum K.E

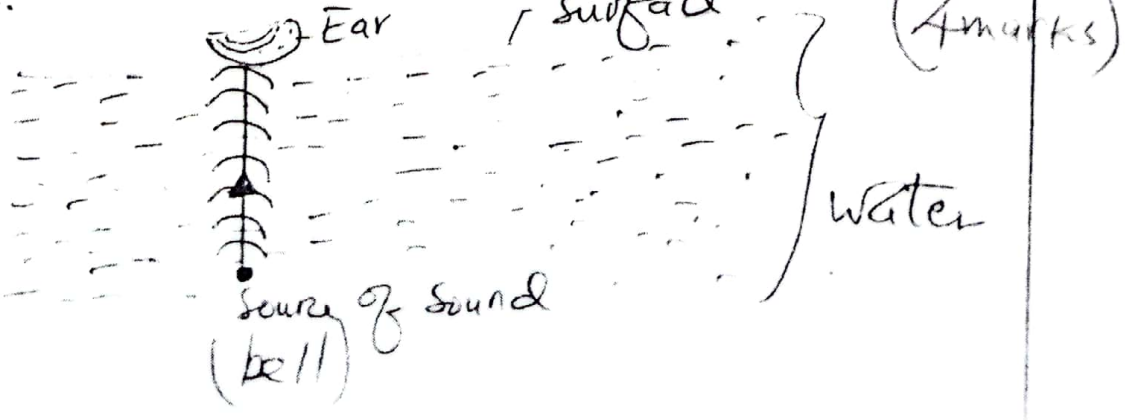
$= \frac{1}{2}mV^2$

$= \frac{1}{2} \times 0{\cdot}5 \times 2^2$

= 1 Joule.

So: potential energy of a Compressed spring is 1 Joule. (2marks)

4 (a) Keep one of your ears gently on the surface of water without letting water into the ear. Now ring the bell inside the water. You will be able to hear the sound clearly. This shows that sound can travel through water.



(4marks)

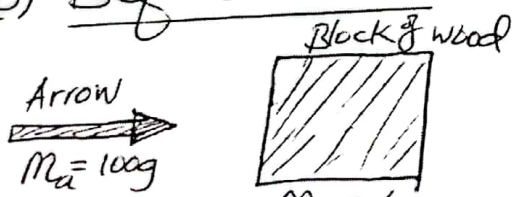
5. (a)(i) - Oscillation of a simple pendulum.
- A ball/an object thrown up ~~air~~ in air.
- Falling of a fruit from a tree
- Slidding down a hill/mountain.

(2marks) any two.

(ii) Elastic collision	Inelastic collision
- Both K.E and momentum are conserved	- Only momentum is conserved
- Mechanical energy is not dissipated	- Mechanical energy is dissipated into heat, light, sound etc
- Forces involved are conservative forces	- Forces involved are non-conservative forces

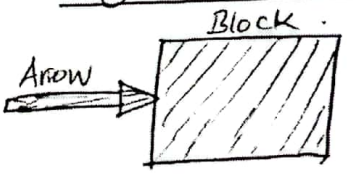
(4marks)

(b) Before collision



$m_a = 100g$
$u_a = 15m/s$
$m_b = 400g$
$u_b = 0m/s$ (rest)

After collision



$V_c = ?$ (Common velocity)

From the principle of conservation of linear momentum:

Total momentum before = Total momentum after impact:

$$m_a u_a + m_b u_b = (m_a + m_b) V_c$$

($u_b = 0$)

$$V_c = \frac{m_a u_a}{m_a + m_b} = \frac{100g \times 15m/s}{100g + 400g}$$

$$V_c = \frac{100g \times 15 m/s}{500g} = 3 m/s$$

(4marks)

∴ The common velocity after impact is 3m/s.

6. (a) It is dangerous to connect the switch in the neutral wire in case of excessive current The fuse blows off due to excessive current and the current stops to flowing in the circuit. But the appliance still remains connected to the high potential point of the supply through the live wire. Now if a person touches the body of the appliance he/she gets an electric shock. Thus, it is highly unsafe to connect a fuse (switch) in the neutral wire. (2marks)

(b)(i) 1 - Earth (E)
2 - Neutral (N) (3marks)
3 - Live (L)

(ii) Terminal 1 is connected to the outer metallic case of the appliance. (3marks)

(iii) The fuse is connected to live wire joined to 3 because in case of excessive flow of current fuse melts first and breaks down the circuit to protect appliances. (2 marks)

7 (a)(i) When the bulb is lighted, the levels of oil in limb X falls while in limb Y rises. (2marks)

(ii) On lighting the bulb, The bulb which is painted black absorbs radiant energy (heat), as a result the gas inside the bulb expands pushing ~~the oil~~ down the oil in limb X in turn causes a rise of oil in limb Y. (6marks)

(b) Atmospheric pressure is low at high altitudes. Hence, boiling point of water decreases and so it does not provide the required heat energy to its contents for cooking. Due to this reason it becomes difficult to cook vegetables on hills and mountains. (2marks)

8. (a) (i) To attain the room temperature, ice cream absorbs heat energy as well as latent heat while water absorbs only heat energy. Thus ice cream absorbs more amount of heat energy from the mouth as compared to water. for this reason, ice cream appears colder than water at 0°c. (2marks)

(ii) Water has ~~high~~ Large value of specific heat capacity, Therefore it absorbs more heat energy from the body at a small temperature change as a body cools. for this reason, water is a preferable liquid for quenching thirst (cooling). (4marks)

Data.

8(a) Mass of water (M_1) = 45g

Temperature of water (T_1) = 50°C

Mass of copper (M_2) = 50g

Temperature of copper (T_2) = 18°C

final temperature (T) = ?

The specific heat capacity of the copper

$C_2 = 0.39 J/g/K$

Specific heat capacity of water $C_1 = 4.2 J/g/K$

$$M_1C_1(T_1 - T) = M_2C_2(T - T_2).$$

$$T = \frac{M_1C_1T_1 + M_2C_2T_2}{M_2C_2 + M_1C_1}$$

$$T = \frac{(45 \times 4.2 \times 50) + (50 \times 0.39 \times 18)}{(45 \times 4.2) + (50 \times 0.39)}$$

$$T = \frac{(9450 + 351)}{(189 + 19.5)} = \frac{9801}{208.5}$$

$T = 47°C$. (3marks)

∴ The final temperature is 47°C.

Assumptions:- We assume no loss of heat energy to the surrounding air. (1mark)

9. (a) This means that a substance is 0.65 times less denser than water. (2.5marks)

(b) Mass of hydrometer = mass of water displaced
= 28g (from the law of flotation)

Volume of Water displaced = $\frac{\text{mass of water displaced}}{\text{density of water}}$

$= \frac{28g}{1g/cm^3} = 28cm^3$

Volume of water displaced = Volume of portion of hydrometer in water.

Volume of hydrometer immersed = $28cm^3$.

Volume of hydrometer above water = Ah
$= 0.75cm^2 \times 3cm$
$= 2.25cm^3$.

(i) Total volume of hydrometer = $28cm^3 + 2.25cm^3$ (.03)
$= 30.25cm^3$.

(ii) When it floats in a liquid of R.D = 1.4

Volume of liquid displaced = $\frac{\text{Mass of liquid displaced}}{\text{Density of liquid}}$

$= \frac{28g}{1.4g/cm^3} = 20cm^3$.

Volume of hydrometer immersed = $20cm^3$.

Volume of stem above the liquid = $(30.25 - 20)cm^3$
$= 10.25cm^3$

Length of stem above, from V = Ah

$10.25cm^3 = 0.75cm^2 \times h$

$$h = \frac{10.25\,cm^3}{0.75\,cm^2} = 13.67\,cm. \quad (3marks)$$

∴ The length of stem above the surface when it floats in a liquid of R.D 1.4 is 13.67cm.

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9. (c) Volume of cork $= 100cm^3$

Density of cork $= 0.25g/cm^3$

Mass of cork = Density of cork × Volume of cork

$= 0.25g/cm^3 \times 100\,cm^3$

$= 25g$.

From the Law of flotation

Mass of cork = Mass of water displaced (25g).

Mass of water displaced $= 25g$.

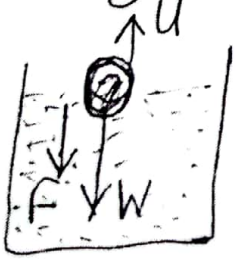
$$\text{Volume of water displaced} = \frac{\text{Mass of water displaced}}{\text{Density of water}}$$

$$= \frac{25g}{1g/cm^3}$$

i) Volume of water displaced $= 25cm^3$.

∴ Volume of cork immersed $= 25cm^3$. (1mark)

ii)



Upthrust = weight of water displaced

$= mg$ ~~$= 0.25g \times 10N/g$~~

$= 0.025kg \times 10$

$= 0.25N$ (Weight of cork)

If it is totally immersed volume of water displaced $= 100cm^3$, mass $= SV = 1g/cm^3 \times 100cm^3$

mass of water displaced $= 100g$

Weight of water displaced when totally immersed
$= mg = 0.1kg \times 10N/kg$
$= \underline{1N}$.

Total upthrust $= W + F_{down}$

$1N = 0.25N + F_{down}$.

$F_{down} = 0.75N$.

The force required to immerse the cork completely is 0.75N. (3marks)

10. (a) Recording studios and concert auditoriums sometimes have their ceilings and walls covered with soft perforated boards so as to absorb sound waves from the source there by preventing from being reflected as echoes from the walls. This enables sound to be heard clearly. (2.5marks)

(b) Data:
Third overtone (4th harmonic) = 840Hz
Velocity of sound, $V = 330m/s$
For open pipe, $f_n = (n+1) f_0$ for overtones.

For 3rd overtone, $n = 3$

$f_3 = (3+1) f_0$, $840 = 4 f_0$.

$f_0 = 210Hz$.

i/ From $f_0 = \frac{V}{2l}$ for open pipe.

$210 = \frac{330}{2l}$, $l = \frac{330}{2 \times 210}$

$$l = \frac{330}{420}\ m = 0.79\ m.$$

$l = 0.79$m or 79cm.

The length of the open pipe = 0.79m or 79cm. (3marks)

(ii) fundamental frequency, $f_0 = 210$Hz. (3marks)

(c) Data:

$f_1 = 384$ Hz

$T_1 = T$

$f_2 = ?$

$T_2 = T + \frac{2}{100}T = 1.02T$.

From $\frac{f_1}{f_2} = \sqrt{\frac{T_1}{T_2}}$

$$\frac{384}{f_2} = \sqrt{\frac{T}{1.02T}}$$

$$f_2 = \sqrt{1.02} \times 384\ Hz$$

$= 388$ Hz.

Beat $= f_2 - f_1 = 388Hz - 384Hz$.

$= 4Hz$. (4marks)

11. (a)

Human eye	lens Camera
Lens: - Biological - flexible	- Lens is artificial . a rigid glass or plastic
Focal length: Variable	For camera is fixed.
Distance: The distance between the lens and retina is fixed.	The distance btn the film and the lens is variable.
Focusing: By changing the shape of the lens	By moving the lens relative to the film.
Aperture: Controlled by the Iris.	Controlled by the diaphragm
Exposure: Is continuous	Controlled by the shutter
Light sensitive surface: ~~Film~~ Retina	film.

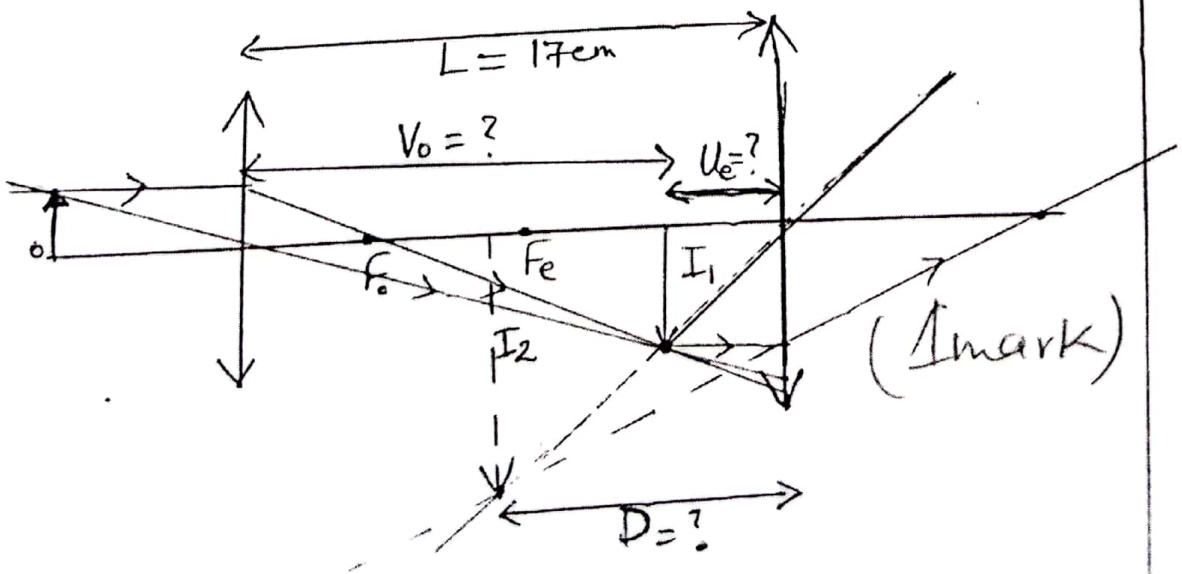
any five (5marks)

(b) Shutter controlls the amount of light entering the camera by the length of times the shutter is open. It act as a gate (2·5marks)

(c) Data:

$f_o = 2\,cm$, $f_e = 6\,cm$, $U_o = 2{\cdot}4\,cm$, $L = 17\,cm$

(i)



(1mark)

Refraction from objective lens:

from: $\frac{1}{f_o} = \frac{1}{u} + \frac{1}{v}$, $\frac{1}{2} = \frac{1}{2.4} + \frac{1}{v}$

$v = 12\text{cm}$.

$V_o = 12\text{cm}$ but $V_o + U_e = L$

$U_e = L - V_o = 17\text{cm} - 12\text{cm} = 5\text{cm}$.

$U_e = 5\text{cm}$.

Also Consider the refraction from the eye piece lens:

$\frac{1}{f_e} = \frac{1}{u} + \frac{1}{v}$ but $V = D$.

$\frac{1}{6} = \frac{1}{5} + \frac{1}{D}$, $D = -30\text{cm}$

$D = -30\text{cm}$ (According to R.P the final image is virtual).

The distance of the final image from the eyepiece lens is 30cm. (2marks)

(ii) Linear magnification (Total magnification):

$M = M_o M_e = \left(\frac{V_o}{f_o} - 1\right)\left(\frac{D}{f_e} - 1\right)$

$= \left(\frac{12}{2} - 1\right)\left(\frac{30}{6} - 1\right)$

$= (6-1)(5-1)$

$= 5 \times 4$ (2marks)

$= \underline{\underline{20}}$.